

Experiment No: 07

Experiment: Construction of Phase diagram for a Phenol- Water system

Theory:

Some liquids are miscible in all proportions but in the case of a few others, the solubility of one in the other is limited. Generally, both liquids become increasingly soluble in the other with increasing temperature until a critical solution temperature (above which they become miscible in all proportions) is reached. This is the case with upper critical solution temperature.

At any temperature below the upper critical solution temperature, the compositions of the two liquid layers in equilibrium are fixed and independent of the relative amounts of the two phases.

Procedure:

- Mass out 2.0 cm^3 of pure phenol, accurately into a boiling tube, Add exactly 1.0 cm^3 of distilled water from a burette. Immerse the tube in water bath and heat slowly, initially the system appears turbid. Keep stirring, note the temperature at which the turbidity disappears and it becomes clear.
- Take the tube out of the water bath and allow cooling in air slowly, keeping stirring, note the temperature at which the turbidity appears. The average of the readings is called the miscibility temperature for that composition.
- Add 1.0 cm^3 of distilled water from the burette to the tube repeat the entire procedure. Again repeat for 3, 4, 5, 6, 7, 8, 10 and 12 cm^3 of water compositions or until the complete miscibility has fallen below 60°C .

Calculation:

Mass of phenol = g

Mass of water/ (g)	Percentage by mass of phenol	Miscibility temperature/ $(^\circ\text{C})$		
		Heating	Cooling	Average

- Plot the mutual solubility curve with temperature on the Y- axis and composition on the X-axis.